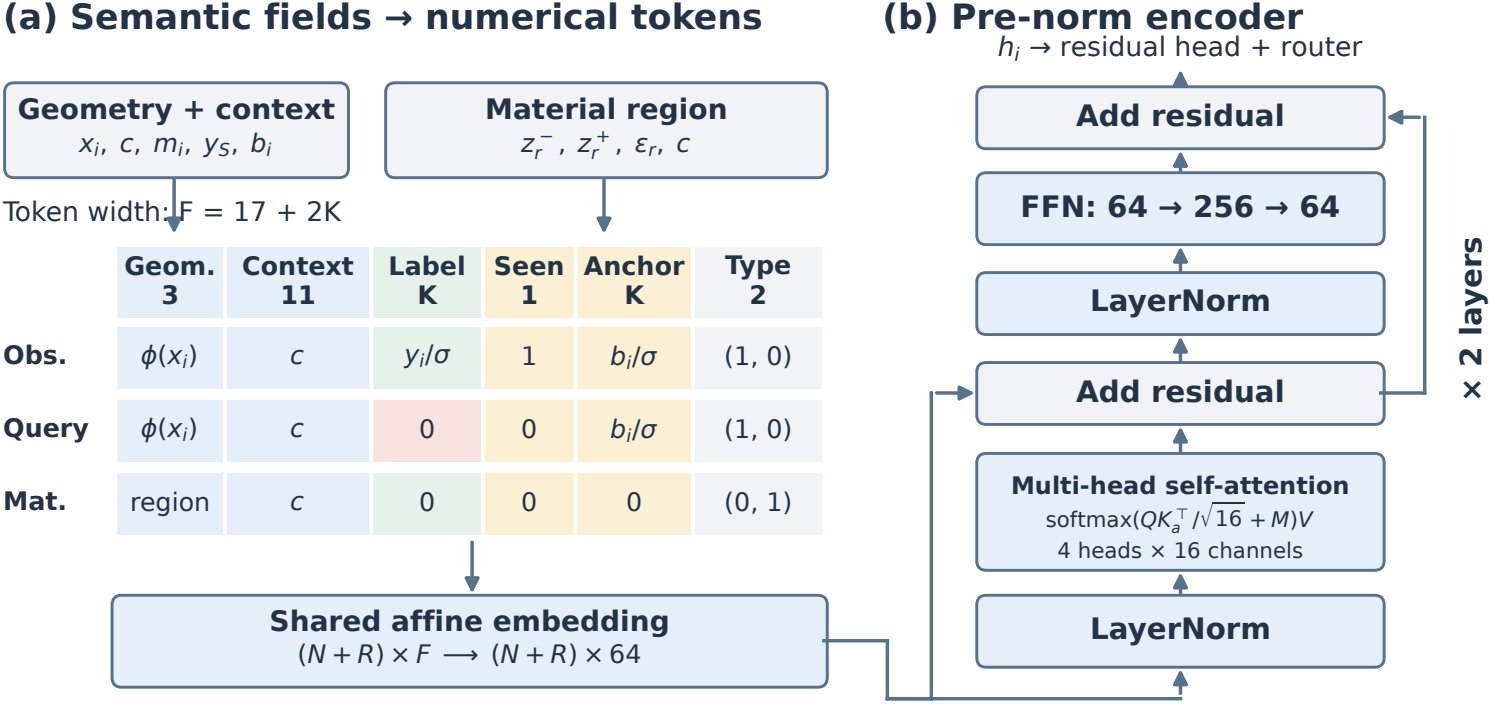
(a) Semantic fields → numerical tokens
Geometry + context
x_i, c, m_i, y_S, b_i
Material region
z_r^-, z_r^+, ε_r, c
Token width: F = 17 + 2K
Geom. 3
Context 11
Label K
Seen 1
Anchor K
Type 2
Obs.
$\phi(x_i)$
c
y_i/σ
1
b_i/σ
(1, 0)
Query
$\phi(x_i)$
c
0
0
b_i/σ
(1, 0)
Mat.
region
c
0
0
0
(0, 1)
Shared affine embedding
$(N+R) \times F \longrightarrow (N+R) \times 64$
Coordinates encode position; padded keys are masked.
(b) Pre-norm encoder
h_i → residual head + router
Add residual
FFN: 64 → 256 → 64
LayerNorm
Add residual
Multi-head self-attention
$\mathrm{softmax}(QK_a^\top/\sqrt{16}+M)V$
4 heads × 16 channels
LayerNorm
× 2 layers